Embodied Movement and Generative Media Performance

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Figure 1: Photographic still of the installation performance

Abstract

This paper documents an interactive dance performance installation in an 8' cubic mesh projection volume. The work presents the difficult history of Asian American immigration and the detention facility on Angel Island in San Francisco Bay. Through a motion capture system, the performer interacts with media projections captured from the detention center, the waterways of the Bay, and from historic archives of the Chinese American immigrant community of San Francisco. Each chapter of the performance engages new movements and poses to create varied generative media conditions.

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CCS Concepts

• **Computer systems organization** → **Real-time systems**; • **General and reference** → *Experimentation*; • **Human-centered computing** → *Interaction design process and methods*; • **Computing methodologies** → *Mixed / augmented reality*; *Image processing*.

Keywords

Dance, Interaction Design, Projection Mapping, Motion Capture

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1 Introduction

Motion capture technology has become increasingly accessible and affordable over the past decade. The discipline of dance studies has utilized these technologies largely in the research, identification and cataloging of cultural dance forms (Protopapadakis 2018). However, the integration of motion capture within real-time dance performance is still uncommon especially outside of high-budget specialty companies. Advances in neural network models have recently lowered the cost of employing these methods (Ding et al 2024) providing opportunities to utilize inexpensive real-time motion capture in a theatrical space. This paper documents an interactive dance performance installation taking place in an 8' cubic mesh projection volume. The work is centered on the immigrant detention facility on Angel Island in San Francisco Bay. The performer's motion interacts with and triggers media projections extending the notion of the virtual dancer (Boucher 2011) into a generative media choreographer.

1.1 Dancer Choreographer

Originally from Chengdu, China, Jingqiu Guan is a filmmaker, choreographer, scholar and dancer. With a broad body of media/dance work centered on questions of identity and culture, she brings both a personal and critical cultural perspective to the troubled history of Angel Island. Jingqiu captured the cinematic and photographic media employed in the projections and created the choreography in dialog with the technological motion capture system.

1.2 Related Works

The integration of motion capture in dance performance is informed by and builds upon important precedents in the field. These include "Living Archive" by Merce Cunningham Dance Company, "Pixel" by Compagnie Käfig and the works of Adrien M and Claire B.

2 Technical Specifications

2.1 Cubic Projection Space

This performance takes place within an 8' cubic projection volume. The structure of this form is engineered from anodized aluminum extrusions with locking plastic corner fittings. The friction between plastic inserts and aluminum spans at each corner holds the structure in place and requires only a rubber mallet for assembly and disassembly.

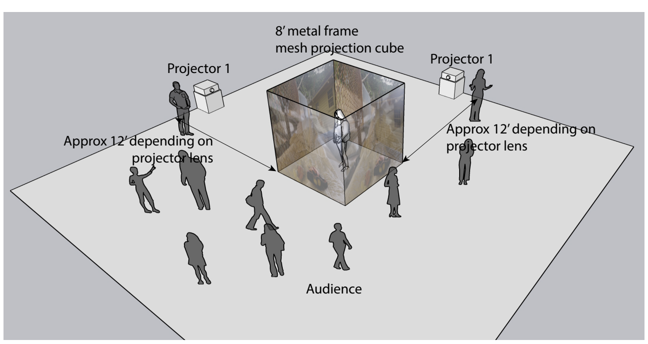

Figure 2: Diagram of performance installation

2.2 Media Projection Surfaces

The four sides of the cubic projection volume are covered in mesh fabric(Figure 2). The rear sides are covered in a ceramic-coated rear projection mesh held in place with grommets and nylon cording. These are the first surfaces to receive light projection from outside of the cube. The remaining two sides of the cube are covered in a white cotton fabric secured in place along the edges of the cube with white tape. These faces are the "audience facing" sides and receive filtered projected light after the rear faces of the cube.

Outside of the cube are two inward-facing projectors. Each projector is calibrated to project fully onto each rear facing 8' square surface. The media passes through the rear projection film and onto the two front facing surfaces. With the dancer performing inside the cube their shadows intersect with this reprojected media creating a variation of focus and scale from the rear projected surfaces to the front surfaces. The dancer within is able to simultaneously view and respond to the media while poetically intertwined and projected with it onto the external, audience-facing surfaces.

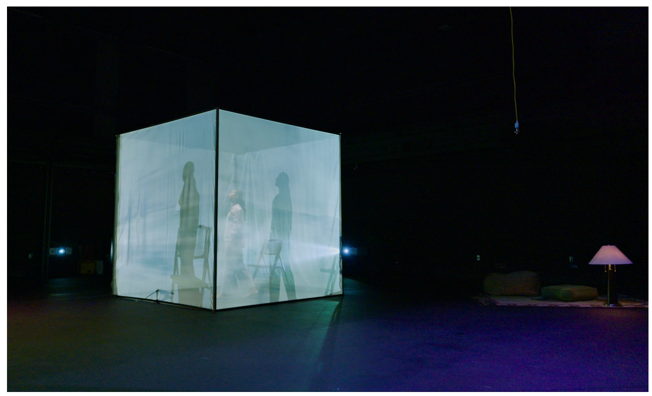

Figure 3: Photograph of the 8' cubic projection volume. The dancer can be seen inside surrounded by projections of ocean water from the first chapter exploring the journey over water to Angel Island.

2.3 Motion Capture System

Inside the cubic volume an inexpensive camera is mounted on the inside bottom front frame. This camera utilizes detachable lenses with C format mounts. Due to the constrained physical volume of the performance space and the location of the camera a wide angle lens was required to maximize capture of the performer. A fish-eye style lens was employed, capturing the dancer through the majority of the constrained performance volume. This camera is connected to the performance control workstation through a standard USB cable routed along the bottom of the cubic volume.

A Python notebook on the control workstation runs a YOLO8 pose analysis model throughout the performance. This model predicts the location of joint locations on each human pose identified in the incoming camera capture. Importantly, the output from the model is an array of poses ordered according to depth. This provides the means for audience members to circulate behind the cube despite being potentially in the camera view. The ordering of the captured poses discards all but the closest pose, the dancer. The captured pose details are broadcast through OSC to the media control system. This separation of camera capture processing from media projection control results in a nodal, scalable system.

2.4 Media Control System

An application developed in TouchDesigner by Derivative Software is used to control the projected media during the performance. The dancer's pose data are used in various combinations throughout the performance to trigger changes in media. Native TouchDesigner nodes are employed for both smoothing and organizing the pose analysis data and for managing projected video output. The motion data entering TouchDesigner from the OSC channels has noise embedded from the sequential output of the YOLO model. Nodes including CHOP Lag and CHOP Math are used to average the data values and allow direct use of these in the transformation of projected media. Native TouchDesigner nodes are also used in the loading, manipulating and output of video data. The DAT Folder and DAT Select nodes are used extensively to allow for dynamic media loading from the control workstation operating system.

Python scripts in the DAT Execute node are used for a large amount of interaction design logic. The ability to create variables, custom functions and to execute consistently at the start of each frame made prototyping complex logic in collaborative workshops efficient and predictable. In addition to triggering changes in projected media the Python scripts are used for custom timers and trigger resets, evaluating the state of the dancer for meeting trigger conditions, debouncing triggers to make interaction predictable, and high level chapter level timers providing means for the performer to reset within the cubic volume.

3 Interaction Design Chapters

The TouchDesigner application was subdivided into four subsections that coincided with content chapters in the performance. During the performance, a technical operator queues changes between sequential chapters, turning on and off specific real-time logic structures.

3.1 Chapter 1: Journey

This performance chapter synthesizes the ocean journey of both the original immigrants from Asia and the performer's research journey to Angel Island. The chapter begins with the performer entering the cubic volume where only a bamboo chair sits. Projected video footage of ocean scapes gradually fades into view as the performer sits. Using the motion capture data of both shoulders the TouchDesigner application assesses the angle of torso rotation, or side to side leaning in the performer. This leaning is transferred to projected video footage of ocean horizons, boat wakes and shorelines. As the performer begins to sway their body to the left and right the horizon on the video footage projected behind them matches. From a restful lulling motion to an energetic frenzied torque the projected footage matches the the performer's movements.

In addition to the rotation of the shoulders, the TouchDesigner application simultaneously tests the height of the shoulders against a calibrated value. When the performer stands high enough to trigger a change the system switches the video projection to still images of fencing, barriers and razor wire. When the performer sits again the footage of ocean horizons and water wakes resumes.

3.2 Chapter 2: Exploring

The second performance chapter presents the exploration of the Angel Island landscape. The chapter begins with fully darkened projectors and tracks both wrists of the performer. When the wrist locations are within a small calibrated distance from each other a bright circle of light brightens the media projection. Within the circle, we see a photograph from the exterior of Angel Island. These images cycle through a folder of grass, tree, beach, and architecture images. The highlight circle follows the wrists of the performer, allowing them to "paint" light across the cube exposing new parts of the photographic images. Gradually the highlight fades back to darkness and a timer resets the system to allow a new triggering.

This simple mechanic allows the performer to move in a wide array of full-body choreography without concern for inadvertently triggering a highlight. By keeping their wrists from crossing they can intentionally utilize this triggering relationship only when they desire direct interaction with the cube walls. Similarly, the computational timer running once a highlight is generated manages a general minimal reset pace to emphasize each highlight as a key moment of media exploration.

Figure 4: Photographic still of the installation performance from Chapter 3: Captivity. The dancer is seen in action with the interrogation chair. Projected photographs of the detention living facility surround her.

Figure 5: Photographic still of the installation performance, Chapter 3

3.3 Chapter 3: Captivity

The third performance chapter presents the violence of captivity and interrogation on Angel Island. (Figure 4, 5) The performer interacts again with the single chair, alternating between subjugation by and violent response to the dominant power structure imposed on the site. The projected imagery in this chapter are modern photographic images of the detention center interior at Angel Island. High-density bunkbeds, personal artifacts and contained spaces of interrogation are all featured. With the increased velocity and violence of choreography in this chapter the interaction system was pushed past its limits. The motion capture system samples frames from the camera on a workstation that also shares projected media management. In addition, the pose analysis model executes on the GPU hardware resources of the same shared workstation. It became apparent in early tests that predictably referencing multiple joint locations in this chapter was unattainable due to the speed and velocity of the performer's choreography. The inability of the pose analysis model to execute at pace with the violent performance movements created an unreliable control mechanism. Over time this chapter's interaction settled on a single motion trigger when the performer's torso crossed a high calibration point in the upper box. Crossing this point, which required standing on the chair, became a specific performance movement that was both calculatable and specific enough to avoid inadvertent triggering.

3.4 Chapter 4: Resistance

The final chapter of the performance references a series of poetic texts recently discovered at the Angel Island facility. Over time the detained immigrants had traced these carvings, gradually embedding them deeply into the plaster wall surfaces. The imagery projected in the performance is a series of animated poetic texts extracted from site photographs. The animation was limited to a simple fade-in of each text and the control system both located them on the projection surface and triggered the visibility of each to fade in. It was important that the performer play the role of writing within the space, referencing the cubic projection wall surfaces while exposing and deepening the found texts. A series of nodes in TouchDesigner were calibrated to follow the performer's right hand and to test for reciprocal back and forth movement of the hand. This system allowed the performer to select locations and times within the chapter to inscribe the poems through a vigorous back and forth oscillation of their hand. Following the projection of six poetic texts a collection of historic portraits covered the projection surfaces. These images of Asian American immigrants from the late 19th and early 20th century were obtained from the Angel Island Archive.

4 Sound Design

Projected filmed footage from Angel Island and San Francisco Bay featured sounds of water and blowing wind. A musical score composed and performed by Karam Salem played throughout the performance. Given the generative nature of the projected visuals, the score provided a consistent tone and timing for each chapter in the work. This consistency aided in the choreography of the performer and the transitions between chapters in the projection design software.

5 Reflections

5.1 On Choreography and Interaction Design

This installation was the result of a collaboration between a dancer and a media artist. Over the course of five months the work was developed from an initial concept diagram into a 16 minute installation performance. The practical discussion, experimentation and programming of the interaction design system was both a notably challenging and rewarding component of the work. The creative bridge between disciplines occurred where embodied movement, technological tracking and media phenomenology crossed paths. The chapter based orientation of the work not only structures the narrative but simultaneously frames limits, opportunities and modalities of interaction explored over the short development timeline of the work.

5.2 Future work

A revised and expanded form of this work will be presented in the spring of 2025 in Los Angeles, California. The specific embodied motions used for triggering media in several chapters are being re-engineered to better accommodate the longer and more varied choreography. The role of the control workstation technician is expanding to trigger specific subchapter conditions within the performance. While the original system was meant to be fully automated it became clear in the performance that this imposed deep limitations on the poses available for consistent capture. The evolving nature of motion capture pose models suggests that we revisit this component to assess new resolutions and outcomes. The current implementation of the YOLO8 pose analysis model is restricted to a 2 Dimensional matrix of joint locations and emerging camera based systems may offer a 3 Dimensional matrix for more complete motion tracking.

5.3 Conclusion

While this work created a dynamic and poetic performance; there remains work to be done in linking the embodied movements to a meaningful generative media narrative. The large technical development hurdles and new collaborative authorship placed a divide between the choreography and narrative depth. The authors maintain that only through ongoing dance and media research within this performance space will the technology provide appropriate expression for the cultural history within.

Acknowledgments

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References

Marc Boucher. 2011. Virtual Dance and Motion-Capture. *Contemporary aesthetics* 9 (2011).

et al. Ding, Jing. 2024. Research on Human Posture Estimation Algorithm Based on YOLO-Pose. *Sensors* 24, 10 (2024), 3036.

Eftychios Protopapadakis, Athanasios Voulodimos, Anastasios Doulamis, Stephanos Camarinopoulos, Nikolaos Doulamis, and Georgios Miaoulis. 2018. Dance Pose Identification from Motion Capture Data: A Comparison of Classifiers†. *Technologies* 6, 1 (2018 2018), 31. https://login.proxy.lib.duke.edu/login?url=https://www.proquest.com/scholarly-journals/dance-pose-identification-motion-capture-data/docview/2026378221/se-2 Copyright - Copyright MDPI AG 2018; Last updated - 2023-12-05.